

DreamBooth Fine-Tuning Practice Report

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1 Introduction

The subject selected for this experiment is a unique Hatsune Miku-themed figurine. This figurine was chosen because it has a distinct visual identity, including long twintails, a black-and-red upper outfit, a white skirt, and a fan accessory. These features make it suitable for testing whether a diffusion model can learn and reproduce a specific object.

2 Dataset Preparation

I collected a small dataset of images of the same figurine and then filtered the images to improve training quality. After screening the data, I selected 10 training images for the final experiment. These images cover a range of viewpoints, including close-up views, half-body views, front full-body views, back views, and three-quarter views.

To make the data more consistent, I applied light preprocessing. All selected images were converted into 512x512 training images while preserving aspect ratio. The preprocessing step focused on reducing background distractions while keeping the figurine intact, especially the twintails, clothing silhouette, and accessory details.

I also prepared lightweight captions using a shared identifier token, `sksmiku`, together with the class word `figurine`. Example captions include:

- a close-up photo of `sksmiku` figurine
- a full-body front view photo of `sksmiku` figurine
- a full-body back view photo of `sksmiku` figurine

This setup was intended to help the model associate the identifier token with the specific figurine while still exposing it to some viewpoint variation.



Figure 1: Representative training images of the selected figurine.

3 Model and Training Setup

I used Stable Diffusion v1.5 as the base model and performed DreamBooth-style LoRA fine-tuning using the Hugging Face diffusers training pipeline.

The main training settings were as follows:

- Base model: `runwayml/stable-diffusion-v1-5`
- Fine-tuning method: DreamBooth LoRA
- Resolution: 512x512
- Batch size: 1
- Gradient accumulation steps: 4
- Learning rate: `1e-4`
- LoRA rank: 8
- Mixed precision: `bf16`
- Seed: 42

The main identifier prompt was:

- `a photo of sksmiku figurine`

I first trained a baseline model with 1200 steps and then trained a second model with 1400 steps. Initially, I considered comparing 800 and 1400 steps. However, after observing that the 1200-step model still had limited subject fidelity, I concluded that comparing 1200 versus 1400 would provide a more meaningful analysis of whether additional training improved subject learning.

To verify that training progressed normally, I also examined the TensorBoard logs stored in the output directory. The loss curves for both runs show a rapid decrease at the beginning followed by a noisier but generally stable trend, which suggests that optimization was functioning properly.

Setting	Value
Base model	Stable Diffusion v1.5
Method	DreamBooth LoRA
Resolution	512
Batch size	1
Gradient accumulation	4
Learning rate	1e-4
LoRA rank	8
Mixed precision	bf16
Compared training lengths	1200 vs 1400 steps

Table 1: Main training setup used in this project.

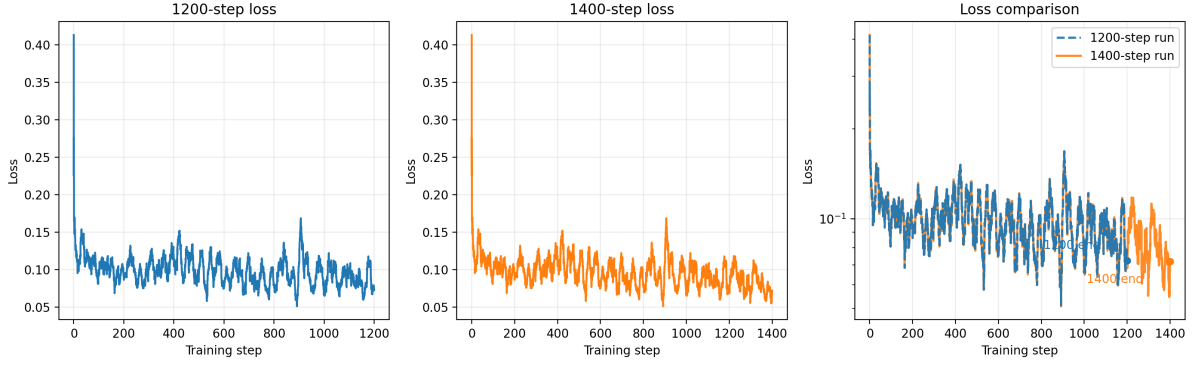


Figure 2: Training loss curves extracted from the TensorBoard logs for the 1200-step and 1400-step runs. Both runs show a strong initial decrease followed by a relatively stable optimization phase.

4 Generated Results

To evaluate the trained LoRA models, I generated images under four types of prompts: basic subject reconstruction, simple scene change, small attribute modification, and light style variation.

Representative prompts include:

- a photo of sksmiku figurine
- a photo of sksmiku figurine, full-body, simple background
- a photo of sksmiku figurine in a cafe
- a photo of sksmiku figurine wearing a small crown
- a photo of sksmiku figurine with an anime-style background

Overall, the generated images show that the model learned some important visual traits of the figurine. In many cases, the model preserved the long twintails, the black-and-red clothing, the white skirt, the fan accessory, and the general figurine-like appearance.

However, the subject fidelity remained limited. The generated outputs often resembled a Miku-themed figurine in general, but they did not always consistently reproduce the exact specific figurine used for training. Facial details, hair ornaments, and local structure were frequently unstable. This indicates that the model learned part of the subject identity, but not the full identity with high reliability.

Prompt fidelity was moderate. The model responded to prompts involving simple new environments, small accessories, and mild style changes. At the same time, stronger prompt variation tended to weaken the subject identity. This suggests that prompt following was partially successful, but still competed with subject preservation.



Figure 3: Representative generated images from the 1200-step model. Suggested examples: basic reconstruction, simple background, scene change, and attribute modification. (Followed prompt above)



Figure 4: Representative generated images from the 1400-step model. Suggested examples should correspond to the same prompt categories as Figure 3 for direct comparison.

5 Comparison Between 1200 and 1400 Steps

The 1200-step model learned the main visual category of the subject, but its outputs often looked like a similar Miku-style figurine rather than the exact trained object. This is more consistent with mild underfitting than with clear overfitting.

The 1400-step model showed a small improvement. The overall silhouette and several major visual elements became slightly more stable, especially in simple prompts. In particular, the figurine-like appearance was somewhat more recognizable.

Nevertheless, the improvement was limited. Even at 1400 steps, facial details and local structures were still unstable, and a few generated outputs were corrupted or invalid. Therefore, increasing the training length helped, but it did not fully solve the subject fidelity issue.

In summary:

- 1200 steps: mild underfitting, weaker subject identity
- 1400 steps: slightly better subject fidelity, but still not fully stable

6 Conclusion and Discussion

This experiment demonstrates that DreamBooth-style LoRA fine-tuning can learn a recognizable portion of a specific subject even from a small dataset. The model successfully preserved some important identity cues and could place the figurine into new prompt conditions.

At the same time, the experiment also highlights the difficulty of learning a highly specific figurine identity from a limited number of training images. The model tended to drift toward a more generic “Miku-like figurine” representation instead of consistently reproducing the exact training subject.

From the perspective of the assignment, the results indicate that subject fidelity was only partially successful. The model learned several recognizable traits of the figurine, but it did not reproduce the exact specific subject consistently across prompts. Prompt fidelity was moderate, as the generated images generally matched the main features requested by the prompts. For instance, prompts involving a crown often produced images containing a crown-like accessory, and prompts specifying a new setting were usually reflected in the generated background or scene. However, as the prompts became more varied or stylized, the subject identity became weaker. Taken together, these observations suggest that the current results are more consistent with mild underfitting than with strong overfitting.

Another important observation is that prompt design itself strongly influenced the evaluation results. In an earlier round of generation, I used more aggressive stylized prompts such as “anime illustration” and other highly creative settings. Under those prompts, the generated images drifted more easily away from the specific figurine identity and became closer to a generic Miku-like anime character. After switching to simpler prompts that preserved the training-style structure, such as `a photo of sksmiku figurine`, the generated results became more suitable for evaluating the assignment goals. This suggests that part of the earlier failure was not caused only by limited training quality, but also by the prompt pushing the model too far away from the training distribution.

More specifically, the later evaluation prompt set was:

- `a photo of sksmiku figurine`
- `a photo of sksmiku figurine, full-body, simple background`
- `a photo of sksmiku figurine in a cafe`
- `a photo of sksmiku figurine wearing a small crown`
- `a photo of sksmiku figurine with an anime-style background`

By contrast, the earlier stylized set used prompts such as:

- `sksmiku figurine on a wooden desk, anime illustration, clean line art, soft lighting`
- `sksmiku figurine in a cherry blossom street, anime style, vibrant colors, detailed background`
- `sksmiku figurine as a magical girl, anime illustration, sparkles, pastel colors`

If I were to continue the project, I would likely improve the results by:

- adding more high-quality training images with cleaner and more diverse viewpoints
- increasing training steps slightly further while monitoring for overfitting
- using a caption-aware LoRA pipeline that makes fuller use of per-image caption information